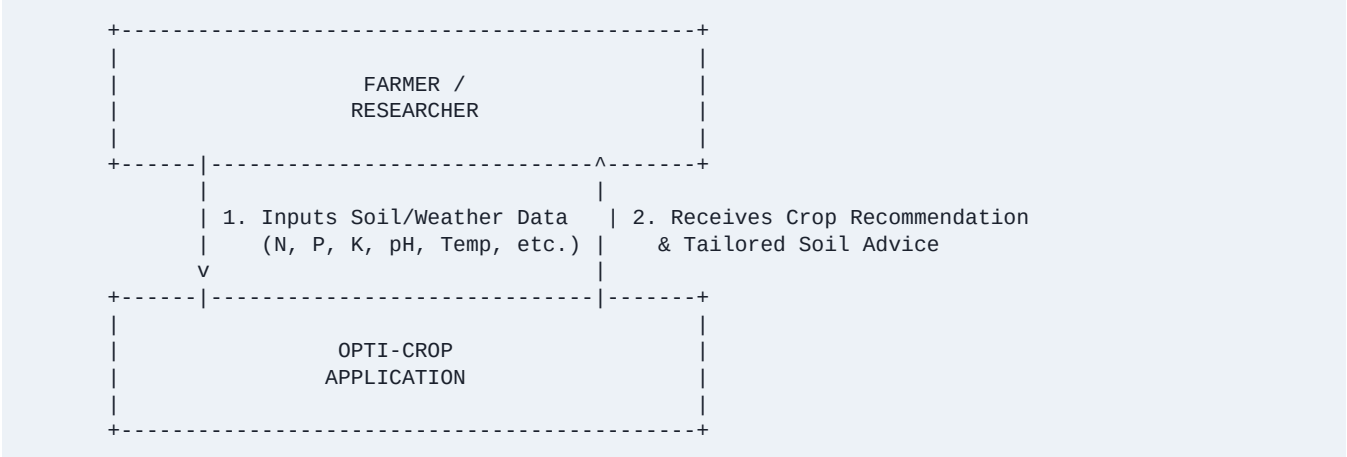


OPTI-CROP PROJECT DELIVERABLE

Data Flow Diagram (DFD)

1. Level 0 DFD (System Context Diagram)

The Context Diagram represents the boundaries of the Opti-Crop system, showing inputs from users and outputs generated by the application.



2. Level 1 DFD (Process Level Diagram)

The Level 1 DFD details how data flows between UI processes, the Machine Learning inference engine, and the SQLite storage database.

1. Process 1.0: User Management

- User inputs credentials (Register/Login).
- Verifies credentials against the **users** table in **opticrop.db**.
- Establishes user session.

2. Process 2.0: ML Prediction Engine

- Receives soil features vector (1x7 array).
- Loads trained model (**model.pkl**).
- Executes prediction -> outputs crop label.

3. Process 3.0: Advice & Analytics Compiler

- Compares user inputs with optimal parameters loaded from **crop_ranges.pkl**.
- Queries **crops** database table for water and seasonal details.
- Computes nutrient deviations and drafts chemical/organic remediation advice.

4. Process 4.0: Log Storage & History

- Saves inputs to **soil_data** table.
- Saves prediction to **predictions** table.
- Saves summary to **reports** table.

- Updates user dashboard history.